

Employee Expense & Travel Management Portal - P1

Objective: Develop a Flask-based corporate expense management system where employees can submit business expenses and travel claims while managers and finance teams review and approve reimbursements.

User Roles

Employee – Create travel requests and submit expenses

Manager – Review and approve employee claims

Finance Admin – Verify expenses and process reimbursements

System Admin – Manage users, policies, and expense categories

Core Features

Secure login and role-based authorization

Employee profile management

Business travel request

Travel approval workflow

Expense claim creation

Expense categories:

Accommodation

Transportation

Meals

Flight

Other business expenses

Receipt attachment

Claim submission

Manager approval/rejection

Finance verification

Reimbursement status tracking

Expense history

Expense limits based on company policies

Dashboard and expense summaries

Search and filter claims

File Handling

Upload invoices and receipts

Support PDF/image attachments

Validate file type and size

Allow finance users to download supporting documents UI Templates

Create Jinja2 pages for:

Login

Employee dashboard

Travel request

Expense submission

My claims

Claim details

Manager approval dashboard

Finance dashboard

Reports Database

Suggested tables:

users

employees

travel_requests

expense_claims

expense_items

expense_categories

expense_receipts

approval_history

reimbursements

expense_policies

Testing Requirements

Test: Authentication and authorization

Travel request workflow

Expense

CRUD

Expense limit validation Approval/rejection workflow

Receipt upload

Unauthorized document access

Reimbursement calculations

Policy validation

Database operations

API error handling

Expected Outcome

A realistic corporate expense management system demonstrating multi-level approval workflows, business-rule validation, Flask APIs, MySQL, authentication/authorization, file handling, Jinja2 templates, and automated testing.

Project 1: Employee Expense & Travel Management Portal (P1)

1. Objective

Develop a **Flask-based corporate Expense & Travel Management System** where employees can submit business travel requests and expense claims, while managers and finance teams review, approve, verify, and process reimbursements.

2. User Roles

Employee

- Create business travel requests
- Submit expense claims
- Upload receipts/invoices
- Track claim and reimbursement status
- View expense history

Manager

- Review employee travel requests
- Review expense claims
- Approve or reject claims
- Add approval/rejection comments

Finance Admin

- Verify approved expenses
- Review supporting documents
- Calculate reimbursements
- Process reimbursements
- Track payment/reimbursement status

System Admin

- Manage users
- Manage expense categories
- Manage company expense policies

- Manage system-level configurations
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3. Core Features

3.1 Authentication & Authorization

- Secure user login
 - Logout
 - Password protection/hashing
 - Role-based authorization
 - Access control based on user role
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3.2 Employee Profile Management

Employees should be able to:

- View their profile
 - Update profile information
 - View their department/designation
 - View their manager
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3.3 Business Travel Management

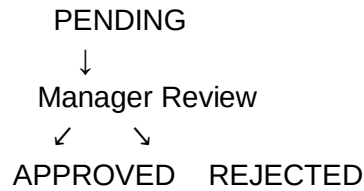
Employees can create travel requests containing:

- Destination
- Travel date
- Return date
- Purpose of travel
- Estimated travel cost
- Additional description

Travel Workflow

Employee creates request





3.4 Expense Claim Management

Employees can create expense claims for business expenses.

Expense Categories

- Accommodation
- Transportation
- Meals
- Flight
- Other Business Expenses

Employees should be able to:

- Create an expense claim
 - Add expense items
 - Edit expense items
 - Delete expense items
 - Calculate total expense
 - Submit the claim
 - View claim history
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3.5 Receipt & Invoice Management

Employees can attach supporting documents to expenses.

Supported Files

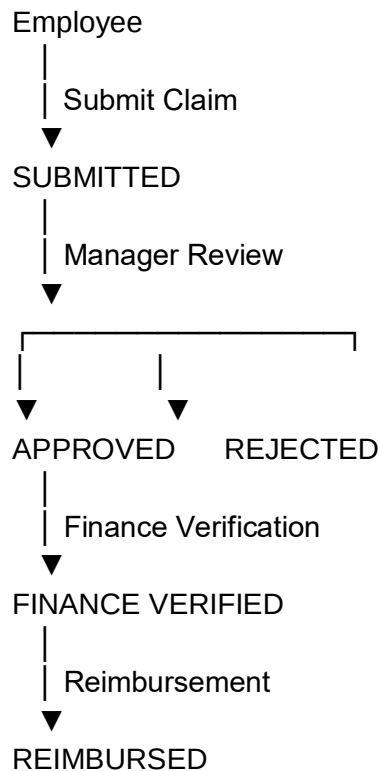
- PDF
- JPG
- JPEG
- PNG

File Handling Requirements

- Validate file type
 - Validate file size
 - Securely store uploaded files
 - Associate receipts with expense items
 - Allow authorized finance users to download documents
 - Prevent unauthorized users from accessing documents
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4. Expense Approval Workflow

The complete expense workflow should be:



Managers should be able to:

- Approve claims
- Reject claims
- Add comments

Finance administrators should be able to:

- Verify expenses
- Review receipts

- Approve reimbursement
 - Process payment
 - Update reimbursement status
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5. Expense Policy Management

The system should support company-defined expense limits.

Example:

Category	Maximum Allowed
Meals	₹1,500/day
Accommodation	₹5,000/day
Transportation	₹2,000/day
Flight	₹15,000
Other	Company-defined

When an employee submits an expense, the system should validate it against the applicable company policy.

Example:

Meal Expense = ₹2,000

Allowed Limit = ₹1,500

Result:

Expense exceeds company policy limit.

6. Dashboard & Reporting

Employee Dashboard

Display:

- Total claims
- Pending claims
- Approved claims
- Rejected claims
- Reimbursed amount
- Recent expenses

Manager Dashboard

Display:

- Pending travel requests
- Pending expense claims
- Approved claims
- Rejected claims
- Employee claim summaries

Finance Dashboard

Display:

- Claims awaiting verification
- Verified claims
- Pending reimbursements
- Completed reimbursements
- Total reimbursement amount

Reports

Support:

- Expense summaries
- Expenses by category
- Expenses by employee
- Expenses by date
- Pending claims
- Rejected claims
- Reimbursed amounts

7. Search & Filtering

Users should be able to search and filter claims based on:

- Employee
- Claim ID
- Expense category
- Claim status
- Date
- Amount

8. Jinja2 Templates

Create Jinja2 pages for:

Login
Employee Dashboard
Employee Profile
Travel Request
My Travel Requests
Expense Submission
My Claims
Claim Details
Manager Approval Dashboard
Finance Dashboard
Reports

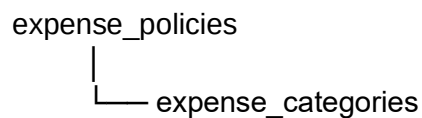
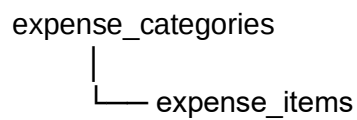
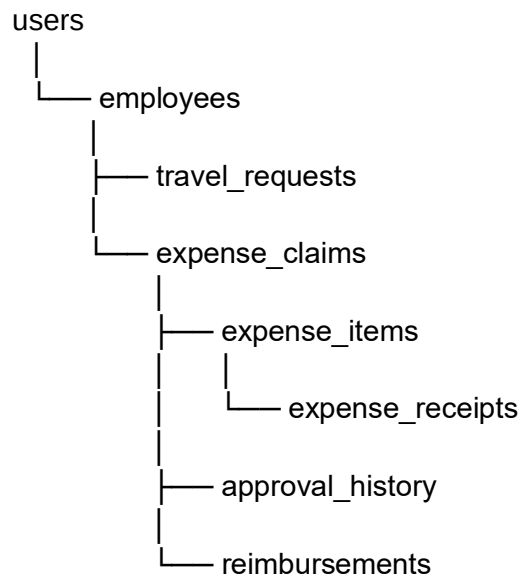
9. Database Design

The suggested database tables are:

users
employees
travel_requests
expense_claims

expense_items
expense_categories
expense_receipts
approval_history
reimbursements
expense_policies

Main relationships



10. Testing Requirements

The application should include automated tests for:

Authentication & Authorization

- Valid login
- Invalid login
- Logout

- Role-based access
- Unauthorized access

Travel Request Workflow

- Create travel request
- View travel request
- Manager approval
- Manager rejection

Expense CRUD

- Create expense
- Read expense
- Update expense
- Delete expense

Expense Validation

- Expense limit validation
- Invalid category
- Invalid amount
- Policy validation

Approval Workflow

- Employee submission
- Manager approval
- Manager rejection
- Finance verification
- Reimbursement processing

File Handling

- Valid PDF upload
- Valid image upload
- Invalid file type
- File size validation
- Unauthorized document access

Reimbursement

- Reimbursement calculation
- Reimbursement status updates

- Payment processing

Database

- CRUD operations
- Foreign-key relationships
- Invalid/missing records
- Transaction handling

API Error Handling

Test appropriate responses for:

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

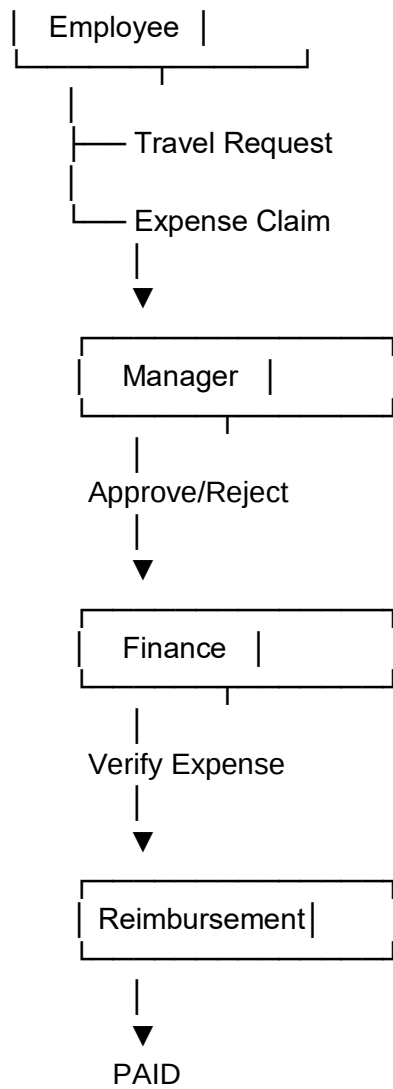
500 Internal Server Error

11. Expected Outcome

The final application should be a **realistic corporate Expense & Travel Management System** demonstrating:

- Flask web development
- Flask APIs
- MySQL database
- SQLAlchemy ORM
- Authentication
- Role-based authorization
- Multi-level approval workflows
- Business-rule validation
- Expense policy enforcement
- File upload and handling
- Jinja2 templates
- Dashboard and reporting
- Search and filtering
- Reimbursement processing
- Automated testing

Overall Workflow



This gives you a clear **Project P1 specification** that you can use as the reference while building each layer.

ER DIAGRAM:

